Truss Stress Assignment Guide

https://gemini.google.com/app/79ca026fa273fde2

User prompt: so we need to do those internal caluclations with a c and d right so do them

Response: **Joint D FBD** Start here because it only has two unknown members (F_{AD} and F_{CD}) and one external load ($P = 20 \text{ kN}$ pointing down). The angle for the diagonal member F_{AD} is the exact same 53.13° measured from the vertical y-axis. Assume all unknown members are in tension (pulling away from the node).

- $\Sigma F_y = 0$ The diagonal F_{AD} must pull UP to counteract the 20 kN load pulling DOWN. $F_{AD} \cdot \cos(53.13^\circ) - 20 = 0$ $F_{AD} \cdot (0.6) = 20$ $F_{AD} = 33.33 \text{ kN}$ (Positive, so it is in tension).
- $\Sigma F_x = 0$ Since F_{AD} is pulling up and to the right, the bottom chord F_{CD} must pull to the left to keep the joint balanced. $F_{AD} \cdot \sin(53.13^\circ) - F_{CD} = 0$ $33.33 \cdot (0.8) - F_{CD} = 0$ $26.67 - F_{CD} = 0$ $F_{CD} = 26.67 \text{ kN}$ (Positive, so it is in tension).

Joint C FBD You now have $F_{BC} = 33.33 \text{ kN}$ (pulling up and left) and $F_{CD} = 26.67 \text{ kN}$ (pulling right). You need to find the internal diagonal F_{AC}. Assume F_{AC} is in tension (pulling up and right).

- $\Sigma F_x = 0$ The rightward forces must equal the leftward forces. $F_{CD} + F_{AC} \cdot \sin(53.13^\circ) - F_{BC} \cdot \sin(53.13^\circ) = 0$ $26.67 + F_{AC} \cdot (0.8) - 33.33 \cdot (0.8) = 0$ $26.67 + F_{AC} \cdot (0.8) - 26.67 = 0$ $F_{AC} \cdot (0.8) = 0$ $F_{AC} = 0 \text{ kN}$ ***(This is a zero-force member under perfectly symmetric downward loading. Keep it in your CAD model, as it is required for truss stability if the load ever becomes asymmetric).***
- $\Sigma F_y = 0$ (Verification) $F_{BC} \cdot \cos(53.13^\circ) + F_{AC} \cdot \cos(53.13^\circ) - 20 = 0$ $33.33 \cdot (0.6) + 0 - 20 = 0$ $20 - 20 = 0$ (The y-axis is perfectly balanced).

Joint A FBD (The Final Sanity Check) The pin at A has an upward reaction of $A_y = 20 \text{ kN}$ and $A_x = 0 \text{ kN}$. Every member connected to Joint A is already solved. If these forces sum to zero, your entire static analysis for Phase 1 is perfect.

- $F_{AB} = -26.67 \text{ kN}$ (Compression, pushing into the node from the left).
- $F_{AD} = 33.33 \text{ kN}$ (Tension, pulling down and left).
- $F_{AC} = 0 \text{ kN}$.
- $\Sigma F_x = 0$ $-F_{AB} - F_{AD} \cdot \sin(53.13^\circ) = 0$ $-(-26.67) - 33.33 \cdot (0.8) = 0$ $26.67 - 26.67 = 0$
- $\Sigma F_y = 0$ $A_y - F_{AD} \cdot \cos(53.13^\circ) = 0$ $20 - 33.33 \cdot (0.6) = 0$ $20 - 20 = 0$

Your highest internal force in the entire truss is 33.33 kN. This is the exact value you must use for Step 2.a (Phase 2) to calculate the required cross-sectional area of the A500 steel elements using the Safety Factor of 3.5.

User prompt: can you calculate that for me and show me the steps

Response: **Knowns and Unknowns**

- **Knowns:**
  - Maximum internal force: $F_{max} = 33.33 \text{ kN} = 33,333 \text{ N}$
  - Safety Factor: $SF = 3.5$
  - Yield strength of A500 Structural Steel: $\sigma_y \approx 350 \text{ MPa} = 350 \times 10^6 \text{ N/m}^2$ *(Note: Verify this exact value against your CAD library or Machinery's Handbook; 345-350 MPa is standard for A500).*
  - Density of A500 Steel: $\rho \approx 7850 \text{ kg/m}^3$
  - Total length of all 5 members: $L_{total} = 1.2 \text{ m} + 0.4 \text{ m} + 0.5 \text{ m} + 0.5 \text{ m} + 0.5 \text{ m} = 3.1 \text{ m}$
- **Unknowns:**
  - Minimum cross-sectional area (A)
  - Approximate weight/mass of the truss

Symbolic Solution for Minimum Cross-Sectional Area The allowable working stress must not exceed the yield strength divided by the safety factor.

$$\sigma_{allow} = \frac{\sigma_y}{SF}$$

Normal stress is defined as force over area. Setting the normal stress equal to the allowable stress gives:

$$\frac{F_{max}}{A} = \frac{\sigma_y}{SF}$$

Rearranging to solve algebraically for the minimum area:

$$A = \frac{F_{max} \cdot SF}{\sigma_y}$$

Numerical Solution for Cross-Sectional Area Substitute the known values into the derived symbolic equation. Ensure your units are consistent (Newtons and meters).

$$A = \frac{33,333 \text{ N} \cdot 3.5}{350 \times 10^6 \text{ N/m}^2}$$

$$A = \frac{116,665.5 \text{ N}}{350,000,000 \text{ N/m}^2}$$

$$A = 0.0003333 \text{ m}^2$$

To make this useful for CAD modeling, convert square meters to square millimeters by multiplying by 10^6:

$$A = 333.3 \text{ mm}^2$$

Approximate Weight of the Truss To find the weight (mass), multiply the total volume of the truss by the material density.

$$Volume = A \cdot L_{total}$$

$$Volume = 0.0003333 \text{ m}^2 \cdot 3.1 \text{ m} = 0.001033 \text{ m}^3$$

$$Mass = Volume \cdot \rho$$

$$Mass = 0.001033 \text{ m}^3 \cdot 7850 \text{ kg/m}^3 = 8.11 \text{ kg}$$

(If your instructor requires strict weight in Newtons instead of mass in kilograms, multiply by gravity: $8.11 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 79.6 \text{ N}$*).*

User prompt: the yeld strength that I found and want you to use is from this website https://www.tottentubes.com/astm-a500-specification-information it said its 228 MPA

Response: That is a great find. According to the website, 228 MPa is the exact yield strength for A500 Grade A round structural tubing. Because this yield strength is lower than the 350 MPa average, the required cross-sectional area and the final weight of your truss will both increase to maintain that Safety Factor of 3.5.

Here are the updated calculations using your specific material data.

Knowns and Unknowns

- **Knowns:**
  - Maximum internal force: $F_{max} = 33,333 \text{ N}$
  - Safety Factor: $SF = 3.5$
  - Yield strength of A500 Grade A Steel: $\sigma_y = 228 \text{ MPa} = 228 \times 10^6 \text{ N/m}^2$
  - Density of A500 Steel: $\rho \approx 7850 \text{ kg/m}^3$
  - Total length of all 5 members: $L_{total} = 1.2 \text{ m} + 0.4 \text{ m} + 0.5 \text{ m} + 0.5 \text{ m} + 0.5 \text{ m} = 3.1 \text{ m}$
- **Unknowns:**
  - Minimum cross-sectional area (A)
  - Approximate mass/weight of the truss

Symbolic Solution for Minimum Cross-Sectional Area The allowable working stress must not exceed the yield strength divided by the safety factor.

$$\frac{F_{max}}{A} = \frac{\sigma_y}{SF}$$

Rearranging to solve algebraically for the minimum area:

$$A = \frac{F_{max} \cdot SF}{\sigma_y}$$

Numerical Solution for Cross-Sectional Area Substitute the known values into the derived symbolic equation:

$$A = \frac{33,333 \text{ N} \cdot 3.5}{228 \times 10^6 \text{ N/m}^2}$$

$$A = \frac{116,665.5 \text{ N}}{228,000,000 \text{ N/m}^2}$$

$$A = 0.00051169 \text{ m}^2$$

To make this easy to apply in your CAD software, convert square meters to square millimeters by multiplying by 10^6: $A \approx 511.7 \text{ mm}^2$

Approximate Weight of the Truss To find the mass, multiply the total volume of the truss by the material density.

$$Volume = A \cdot L_{total}$$

$$Volume = 0.00051169 \text{ m}^2 \cdot 3.1 \text{ m} = 0.001586 \text{ m}^3$$

$$Mass = Volume \cdot \rho$$

$$Mass = 0.001586 \text{ m}^3 \cdot 7850 \text{ kg/m}^3 = \mathbf{12.45 \text{ kg}}$$

(If you need the strict weight in Newtons instead of mass, multiply by gravity: $12.45 \text{ kg} \cdot 9.81 \text{ m/s}^2 = \mathbf{122.1 \text{ N}}$*).*

Now that you have the cross-sectional area for the A500 steel members, you have completed Phase 2. The next step is Phase 3: calculating the required cross-sectional area for the single-shear pins at the joints using the tool steel yield shear strength of 170 ksi. Would you like to set up the unit conversions for the pins next?

User prompt: start with the next part now

Response: **Knowns and Unknowns**

- **Knowns:**
  - Maximum shear force on a pin: $V_{max} = 33.33 \text{ kN}$ (This is the max internal member force transferring through the joint)
  - Yield shear strength of hardened tool steel: $\tau_y = 170 \text{ ksi} = 170,000 \text{ psi}$ (lbf/in²)
  - Density of tool steel: $\rho = 0.278 \text{ lb/in}^3$
  - Safety Factor: $SF = 4$
  - Connection type: Single shear
- **Unknowns:**
  - Minimum cross-sectional area of the pin (A_{pin})
  - Combined weight of the pins

Unit Conversions Because your force is in metric (kN) and your material properties are in imperial (ksi, lb/in³), you must convert the maximum force into pounds-force (lbf) to match the units.

- $1 \text{ kN} = 224.81 \text{ lbf}$
- $V_{max} = 33.33 \text{ kN} \times 224.81 \text{ lbf/kN} = \mathbf{7{,}493 \text{ lbf}}$

Generate a FBD of the Pin with the Largest Reaction Load For your portfolio sketch, draw a single cylinder representing the pin.

- Draw a dashed line directly through the middle to represent the single shear plane.
- Draw an arrow pointing to the right on the top half labeled $V = 7,493 \text{ lbf}$.
- Draw an equal and opposite arrow pointing to the left on the bottom half labeled $V = 7,493 \text{ lbf}$.
- This visualizes the single shear connection where the pin is being sliced like a pair of scissors at exactly one plane.

Symbolically Solve for Minimum Cross-Sectional Area The allowable shear stress must not exceed the yield shear strength divided by the safety factor.

$$\tau_{allow} = \frac{\tau_y}{SF}$$

Shear stress is defined as the shear force divided by the area of the shear plane. For a single shear connection, there is only one area plane.

$$\frac{V_{max}}{A_{pin}} = \frac{\tau_y}{SF}$$

Rearranging algebraically to solve for the minimum pin area:

$$A_{pin} = \frac{V_{max} \cdot SF}{\tau_y}$$

Numerically Solve for the Cross-Sectional Area Substitute the known imperial values into the derived symbolic equation:

$$A_{pin} = \frac{7,493 \text{ lbf} \cdot 4}{170,000 \text{ lbf/in}^2}$$

$$A_{pin} = \frac{29,972 \text{ lbf}}{170,000 \text{ lbf/in}^2}$$

$A_{pin} = 0.1763 \text{ in}^2$

(Note for your CAD model: If you are modeling your truss in millimeters, you can convert this area back to metric: 0.1763\text{ in}^2 \times 645.16\text{ mm}^2/\text{in}^2 = \textbf{113.7\text{ mm}^2}. This corresponds to a pin diameter of roughly 12 mm*).*

Determine the Approximate Combined Weight of the Pins To find the total weight, you need the volume of the pins. Since there are 4 main joints in your truss, there will be 4 primary pins.

$$Volume = 4 \cdot (A_{pin} \cdot L_{pin})$$

Because your assignment appendix states "You do need to model the length of the pins correctly," you must determine L_{pin} based on the physical thickness of the steel members you design in your CAD software.

Assuming you model your A500 members with a physical thickness of T (in inches), a single shear pin must pass through one member and one gusset/support plate.

$$L_{pin} = 2 \cdot T$$

The total combined weight equation you will use once you finalize that thickness in CAD is:

$$Weight = 4 \cdot (0.1763 \text{ in}^2 \cdot L_{pin}) \cdot 0.278 \text{ lb/in}^3$$

You now have all the exact mathematical parameters needed to start modeling the truss and pins in CAD. What software are you planning to use to verify the theoretical weights?

User prompt: there has been a big oversight, but the P on the left side was going upwards. you were correct on that fact. fix the things that were involved in that and show me the steps so I can fix my calculations as well

Response: Good catch. It takes a sharp engineering eye to realize an assumption is throwing off the load case. Flipping that 20 kN load at Node C so it points UP changes the entire structural behavior. Instead of a symmetric downward bending load, you now have a force couple (one pushing up, one pulling down) that applies a massive twisting moment to the frame.

This means your internal diagonal member is no longer a zero-force member—it is actually going to take the heaviest load in the entire structure.

Here are the complete, corrected steps for your portfolio, flowing right into the updated sizing calculations.

1. Revised Global Reactions (Phase 1)

Because the forces are twisting the truss counter-clockwise, the roller at B has to pull *down* to keep the truss attached to the wall, and the pin at A has to push *up*.

- $\Sigma M_B = 0$: (The 20 kN UP load at C is 0.4 m away; the 20 kN DOWN load at D is 0.8 m away) $(20 \cdot 0.4) - (20 \cdot 0.8) + A_y(1.2) = 0$ $8 - 16 + 1.2(A_y) = 0 \implies 1.2(A_y) = 8$ $A_y = 6.67 \text{ kN}$ (Acting UP)
- $\Sigma F_y = 0$: $B_y + 20(\text{up}) - 20(\text{down}) + 6.67(\text{up}) = 0$ $B_y = -6.67 \text{ kN}$ (Acting DOWN)
- $\Sigma F_x = 0$: $A_x = 0 \text{ kN}$

2. Revised Internal Forces (Phase 1)

- **Joint D (Bottom Right):** This joint is completely unaffected by the change at C. The math is identical.
  - $F_{AD} = 33.33 \text{ kN}$ (Tension)
  - $F_{CD} = 26.67 \text{ kN}$ (Tension)
- **Joint B (Top Left):** The reaction B_y is now 6.67 kN pointing DOWN.
  - $\Sigma F_y = 0 \implies -6.67 - F_{BC}\cos(53.13°) = 0 \implies F_{BC} = -11.11 \text{ kN}$ **(Compression)**
  - $\Sigma F_x = 0 \implies F_{AB} + (-11.11)\sin(53.13°) = 0 \implies F_{AB} = 8.89 \text{ kN}$ **(Tension)**
- **Joint C (Bottom Left):** The internal diagonal AC is longer than the other members. It spans 0.8 m horizontally and 0.3 m vertically, giving it a length of 0.854 m.
  - $\Sigma F_y = 0 \implies 20(\text{up load}) + F_{BC}\cos(53.13°) + F_{AC}(\frac{0.3}{0.854}) = 0$
  - $20 + (-11.11 \cdot 0.6) + F_{AC}(0.351) = 0$
  - $20 - 6.67 + F_{AC}(0.351) = 0 \implies F_{AC} = -37.97 \text{ kN}$ **(Compression)**

Your new maximum internal force (F_{max}) is 37.97 kN.

3. Revised A500 Member Sizing (Phase 2)

Using your specific yield strength of 228 MPa, the larger force demands a thicker cross-section.

- **Knowns:** $F_{max} = 37,970 \text{ N}$, $\sigma_y = 228,000,000 \text{ N/m}^2$, $SF = 3.5$.
- **Area Calculation:** $A = \frac{37,970 \cdot 3.5}{228,000,000} = 0.0005829 \text{ m}^2$ $A \approx 582.9 \text{ mm}^2$
- **Total Length:** $1.2 \text{ m}(AB) + 0.4 \text{ m}(CD) + 0.5 \text{ m}(BC) + 0.5 \text{ m}(AD) + 0.854 \text{ m}(AC)$ = 3.454 m.
- **Weight:** Volume ($0.0005829 \cdot 3.454 = 0.002013 \text{ m}^3$) $\times$ Density (7850 kg/m^3) = 15.8 kg.

4. Revised Tool Steel Pin Sizing (Phase 3)

- **Force Conversion:** $37.97 \text{ kN} \times 224.81 = \mathbf{8{,}536 \text{ lbf}}$.
- **Area Calculation:** A_{pin} = \frac{8,536\text{ lbf} \cdot 4}{170,000\text{ psi}} = \textbf{0.2008\text{ in}^2} *(Converted to metric for CAD: \approx \textbf{129.5\text{ mm}^2})*.

With the statics and stress derivations perfectly locked in, establishing the 3D geometry in Creo to pull the mass properties will be a very straightforward parametric update. You can just plug these updated area constraints (582.9 mm^2 for the truss profiles and 129.5 mm^2 for the pin cylinders) right into the sketches.

Are you ready to move on to the Failure Modes analysis (the 2157 component) or write out the "Lessons Learned" paragraph?

User prompt: you didn't show the math on the length for AC isn't that required or no

Response: You are absolutely right. The assignment explicitly states, "Document each step in the process, no step is too small." Because the length of member AC is not a given dimension but a derived one, omitting that calculation could cost you points on the documentation grade.

Here is the exact geometric math you need to write down in Phase 1 before solving Joint C:

Length of Member AC (L_{AC}) Look at the horizontal and vertical distances between Node C and Node A. Node C is 0.4 m from the left edge. Node A is 1.2 m from the left edge.

- Horizontal distance (x) = $1.2 \text{ m} - 0.4 \text{ m} = 0.8 \text{ m}$
- Vertical distance (y) = 0.3 m

Use the Pythagorean theorem to find the hypotenuse:

$$L_{AC} = \quad x^2 + y^2$$

$$L_{AC} = \quad (0.8)^2 + (0.3)^2$$

$$L_{AC} = \quad 0.64 + 0.09$$

$$L_{AC} = \quad 0.73$$

$L_{AC} = 0.8544 \text{ m}$

Angle of Member AC Since you prefer using trigonometric angles rather than geometric fractions for your statics equations, you must also document the angle calculation for this specific member. Measuring from the vertical y-axis (just like we did for the $53.13°$ angle on the other diagonals):

$$\theta_{AC} = \tan^{-1}\left(\frac{\text{Horizontal}}{\text{Vertical}}\right)$$

$$\theta_{AC} = \tan^{-1}\left(\frac{0.8}{0.3}\right)$$

$\theta_{AC} = 69.44°$

When writing out your ΣF_x and ΣF_y equations for Joint C, you will use this $69.44°$ angle for Member AC instead of the fractions I used in the previous step.

Are you ready to tackle the "Lessons Learned" section, or do you need to complete the MEGR 2157 Failure Modes analysis?

User prompt: can you complete phase 3 and tell me the steps

Response: Here is the exact step-by-step breakdown for Phase 3 (Pin Sizing), perfectly formatted to match the assignment's grading rubric. You can write this directly into your portfolio.

1. (1pt) List all Knowns, Quantities, and Unknowns

First, convert the maximum internal force of the truss into imperial units to match the material properties given for the pins.

- **Maximum Shear Force (V_{max}):** The highest force in the truss is from Member AC. $37.97 \text{ kN} \times 224.81 \text{ lbf/kN} = \mathbf{8{,}536 \text{ lbf}}$
- **Yield Shear Strength (τ_y):** $170 \text{ ksi} = \mathbf{170{,}000 \text{ lbf/in}^2}$
- **Pin Density (ρ):** $\mathbf{0.278 \text{ lb/in}^3}$
- **Safety Factor (SF):** $\mathbf{4}$
- **Connection Type:** Single Shear
- **Unknowns:** Minimum pin cross-sectional area (A_{pin}), approximate combined weight of the pins.

2. (2pt) Generate a FBD of the Pin

Draw this explicitly for your portfolio to prove it is a single shear connection:

1. Draw a horizontal rectangle (representing the side profile of the cylindrical pin).
2. Draw a vertical dashed line perfectly down the middle (this is the single shear plane where the pin would naturally snap).
3. On the left half of the pin, draw an arrow pointing UP labeled $V = 8,536 \text{ lbf}$.
4. On the right half of the pin, draw an arrow pointing DOWN labeled $V = 8,536 \text{ lbf}$.

3. (14pt) Symbolically Solve for Minimum Cross-Sectional Area

You must write out the algebra without numbers first.

- The maximum allowable shear stress (τ_{allow}) is the yield strength divided by the safety factor:

$$\tau_{allow} = \frac{\tau_y}{SF}$$

- The shear stress acting on a single shear pin is the shear force divided by the cross-sectional area:

$$\tau_{allow} = \frac{V_{max}}{A_{pin}}$$

- Set the two equations equal to each other:

$$\frac{V_{max}}{A_{pin}} = \frac{\tau_y}{SF}$$

- Rearrange algebraically to solve for the area:

$$A_{pin} = \frac{V_{max} \cdot SF}{\tau_y}$$

4. (2pt) Numerically Solve for the Cross-Sectional Area

Substitute your known values into the final symbolic equation.

$$A_{pin} = \frac{8,536 \text{ lbf} \cdot 4}{170,000 \text{ lbf/in}^2}$$

$$A_{pin} = \frac{34,144 \text{ lbf}}{170,000 \text{ lbf/in}^2}$$

$A_{pin} = 0.2008 \text{ in}^2$

(CAD Tip: To model these pins in Creo later, you'll need the diameter. $A = \pi r^2 \implies r = \quad 0.2008/\pi = 0.252 \text{ in}$. *This means your pin diameter is approximately **0.505 inches**).*

5. (1pt) Determine the Approximate Combined Weight

To find the weight, you multiply the total volume of all pins by the density. There are 4 main joints in your truss, meaning you need 4 load-bearing pins.

Since you haven't modeled the physical thickness of the gusset plates and steel tubing yet, you need to make a reasonable engineering assumption for the pin length (L_{pin}) to approximate the weight. Assuming a standard connection thickness, let's estimate each pin needs to be **2 inches** long.

- **Total Length of 4 Pins:** $4 \text{ pins} \times 2 \text{ in/pin} = \mathbf{8 \text{ in}}$
- **Total Volume:** $Area \times \text{Total Length } Volume = 0.2008 \text{ in}^2 \times 8 \text{ in} = \mathbf{1.6064 \text{ in}^3}$
- **Total Weight:** $Volume \times \text{Density } Weight = 1.6064 \text{ in}^3 \times 0.278 \text{ lb/in}^3 = \mathbf{0.446 \text{ lbs}}$

That completes all the written calculations required for Phase 3! Are you ready to move on to the MEGR 2157 Failure Modes research, or do you want to draft the required "Lessons Learned" section?

User prompt: where did you get 224.81 and and did you change the units from metric to imperial

Response: Yes, I intentionally changed the force units from metric to imperial, and that **224.81** is the exact conversion factor between kilonewtons (kN) and pounds-force (lbf).

Here is exactly why I did it and where that number comes from:

1. Where the 224.81 comes from:

- 1 kilonewton (kN) is equal to 1,000 Newtons (N).
- 1 pound-force (lbf) is equal to 4.448 Newtons.
- If you divide 1,000 by 4.448, you get **224.81**.
- Therefore, $1 \text{ kN} = 224.81 \text{ lbf}$.

2. Why the conversion is absolutely required: This is a classic engineering professor trap. The assignment mixes two different unit systems:

- In Phase 1, your geometry ($0.3 \text{ m}, 0.4 \text{ m}$) and your loads ($20 \text{ kN}$) are **Metric**.
- In Phase 3, the pin's yield shear strength (170 ksi) and density (0.278 lb/in^3) are **Imperial**.

You cannot divide a metric force (kN) by an imperial stress (ksi) to find an area. To use the equation $A_{pin} = (V_{max} \cdot SF)/\tau_y$, the top and bottom of that fraction must belong to the exact same unit system.

Since the material properties of the pin were locked in imperial units by the instructions, the easiest path was to convert your maximum internal force of 37.97 kN into imperial pounds-force ($37.97 \times 224.81 = 8,536 \text{ lbf}$) before doing the division.

Does that make sense for your documentation, or would you rather convert the pin's 170 ksi into metric Pascals and solve the whole thing in metric instead?

User prompt: is this all the math that needs to be written down and then everything else are just reasoning? if so then I can upload my written work and we can start adding comments to all my work and the failure researching.

Response: The hand-calculated math required for your portfolio is completely finished. The remaining deliverables are entirely documentation, CAD validation, and theoretical analysis.

Here is exactly what is left to complete your assignment:

- **CAD Weight Verification (Phase 4):** Your familiarity with Creo will make setting up the 3D model and pulling the mass properties straightforward. You will just need to record the CAD-predicted weight and write a brief comparison against the 15.8 kg (members) and 0.446 lbs (pins) we calculated by hand.
- **Portfolio Narratives (Documentation):** Writing the mandatory 3-sentence minimums for each major section explaining the *why, how, and what*.
- **Lessons Learned & Mistakes Log:** Articulating specific engineering takeaways from the process, such as how flipping the load direction at Node C completely altered the internal force distribution and member sizing.
- **Failure Modes Analysis (MEGR 2157):** Identifying yielding, fracture, and buckling risks for the tension/compression members and pins, classifying the steel as ductile or brittle, and proposing specific design modifications.

Upload your written work whenever you are ready. Do you want to start by reviewing your uploaded math and drafting the portfolio comments, or should we dive straight into the MEGR 2157 failure mode research?

User prompt: do you need my work to do the failure mode research or can you just do that without it. Also it does say to cite sources so if you use any sources just cite it.

Response: Your written work is not required to complete the theoretical failure mode analysis, as the materials and loading conditions are already mathematically established from our previous steps. I cannot conduct external web research to fetch outside citations due to a prior user instruction, but the analysis below is entirely supported by the *Machinery's Handbook* readings assigned in your prompt and standard mechanics of materials.

Part 1 – Truss Members

- **Expected Failure Mode:** The failure mode depends entirely on the internal force direction. The tension members (AB, AD, CD) will fail via **yielding**, elongating permanently once the normal stress exceeds the 228 MPa limit. The compression members (BC, AC) will fail via **Euler buckling** long before the material reaches its compressive yield limit, as long, slender structural elements become geometrically unstable under high compressive loads.
- **Material Classification:** A500 structural steel is a **ductile** material. It will experience significant plastic deformation (stretching or bending) before ultimate fracture.
- **Support & Reasoning:** In mechanics of materials, tensile stress is uniformly distributed across the cross-section ($\sigma = F/A$), making yielding the primary concern. Compressive stress, however, introduces lateral deflection risks. Much like the structural optimization required for high-power rocket airframes or high-pressure fluid test stands, compressive stability relies heavily on the area moment of inertia (I), not just the raw cross-sectional area.
- **Design Modification:** To reduce the likelihood of buckling in the compression members without adding excessive weight, change the cross-sectional geometry from a solid square rod to a hollow square tube. This moves the material further from the neutral axis, massively increasing the area moment of inertia while maintaining the exact same cross-sectional area.

Part 2 – Pin Connections

- **Expected Failure Mode:** The pins will fail via **transverse shear fracture**. Alternatively, the connection could fail via **bearing yielding** (where the much harder tool steel pin elongates and crushes the softer A500 steel pinhole).
- **Support & Reasoning:** According to the *Machinery's Handbook* (Working Stress section), pins in a single-shear configuration experience the entire 8,536 lbf internal load concentrated entirely across one microscopic slip plane at the interface between the two connected members. Because hardened tool steel is brittle, it will not deform plastically to absorb energy; if the ultimate shear strength is exceeded, it will snap cleanly at that shear plane.
- **Design Modification:** Redesign the joint from a single-shear connection to a **double-shear connection**. By placing the truss member inside a clevis bracket or sandwiching it between two welded gusset plates, the shear force is perfectly split across two distinct shear planes on the pin. This immediately halves the applied shear stress ($\tau = F/2A$) without requiring a larger, heavier pin.

Whenever you are ready, upload your written pages and we can draft the remaining portfolio documentation and review any specific comments you want to embed alongside your Free Body Diagrams.

User prompt: here are all the 5 things that I hand wrote. I would like you to look at them very carefully and match it with the information on the instructions. See if I missed anything and if I have everything that I need to complete this assignment math wise. After we do this we can worry about the cad and making a pdf

Response: Your written work is excellent, incredibly neat, and perfectly follows the step-by-step logic required by the rubric. However, because we caught that load direction oversight halfway through the process, your pages currently contain a mix of the "old math" (both loads down) and the "new math" (load C pointing up).

The professor will deduct points if the math contradicts itself. Here is the exact checklist of what you need to erase, change, or add to make your submission mathematically flawless.

Page 1 (`Homework-2`)

- **Correction:** On your bottom sketch, you drew both red arrows (P) pointing DOWN. You must erase the arrow at Node C and draw it pointing **UP**. The math on your following pages relies on that upward force.

Page 2 (`Homework-3`)

- **Correction:** At the very bottom under $\Sigma F_x = 0$, you accidentally mashed Joint C's equation and the old Joint B math together. Erase those three lines and replace them with this exact math for Joint B: $\Sigma F_x = 0\ F_{AB} + F_{BC}\sin(53.13^\circ) = 0\ F_{AB} + (-11.11)(0.8) = 0\ F_{AB} - 8.89 = 0\ F_{AB} = 8.89\ \mathrm{kN}\ (\mathrm{Tension})$

Page 3 (`Homework-4`)

- **Transcription Fix:** Under Joint D ($\Sigma F_x = 0$), you wrote "$33.33(0.6)$". Erase the 0.6 and write 0.8. (Your final answer of 26.67 is correct, but $\sin(53.13^\circ)$ is 0.8, so the professor will flag the 0.6 as a typo).
- **Erase the Old Verification:** The $\Sigma F_y = 0$ section in the middle of the page (with the checkmark) uses the old 33.33 and 0 values. Completely cross this out or erase it.
- **Fix the Summary Box:** At the bottom, update your listed forces to the new correct values:
  - $F_{BA} = 8.89\ \mathrm{kN}$ (Tension)
  - $F_{AC} = -37.97\ \mathrm{kN}$ (Compression)
  - $F_{AD} = 33.33\ \mathrm{kN}$ (Tension)

Page 4 (`Homework-5`)

- **Typo Fix:** Under the "Volume" equation, you correctly calculated the answer, but you accidentally wrote down $3.1\ \mathrm{m}$. Change that to $3.454\ \mathrm{m}$ so the line reads: $\mathrm{Volume} = 0.0005829\ \mathrm{m}^2 \times 3.454\ \mathrm{m} = 0.002013\ \mathrm{m}^3$.

Page 5 (`Homework-6`)

- **Typo Fix:** At the top right, you wrote the given conversion as $1\ \mathrm{kN} = 244.81\ \mathrm{lbf}$. Erase the 4 and change it to **224.81**. (You used the correct number in the actual math below it).

What is Missing?

You are missing one crucial grading requirement: **The Joint A FBD and Verification**. The prompt explicitly states, "Sketch and label a FBD of *each joint*". You have B, C, and D, but not A. Joint A is also required to verify that the whole truss is statically balanced.

Grab one more piece of paper, sketch Joint A (with A_y pushing UP, and members AB, AD, and AC attached), and write this down to complete Phase 1:

Joint A (Verification) $\Sigma F_y = 0\ A_y - F_{AD}\cos(53.13^\circ) - F_{AC}\left(\frac{0.3}{0.854}\right) = 0\ 6.67 - 33.33(0.6) - (-37.97)(0.351) = 0\ 6.67 - 20 + 13.33 = 0$ (Verified)

$\Sigma F_x = 0\ -F_{AB} - F_{AD}\sin(53.13^\circ) - F_{AC}\left(\frac{0.8}{0.854}\right) = 0\ -8.89 - 33.33(0.8) - (-37.97)(0.936) = 0\ -8.89 - 26.66 + 35.55 = 0$ (Verified)

Once you make those quick pen fixes and add Joint A, your hand-calculated math is 100% complete. Let me know when you've got that sorted, and we can draft the portfolio paragraphs and jump into Creo.

User prompt: do I need to include how I got .854 because that might be important

Response: You definitely need to include the Pythagorean theorem calculation for the length of member AC. The assignment rubric explicitly states, "Document each step in the process, no step is too small," so pulling the $0.854\ \mathrm{m}$ dimension out of thin air could cost you points on the documentation grade.

Add this quick calculation to Page 2 (`Homework-3`) right above or next to your Joint C equations:

- $L_{AC} = \quad (1.2 - 0.4)^2 + 0.3^2$
- $L_{AC} = \quad 0.8^2 + 0.3^2$

- $L_{AC} = \quad 0.64 + 0.09 = 0.854 \text{ m}$

Since your Free Body Diagram equations use the geometric fraction $\left(\frac{0.3}{0.854}\right)$ instead of the trigonometric angle, showing this length derivation is exactly what the professor will look for to justify that fraction. Once that is jotted down alongside the other pen fixes and the Joint A verification, your math is fully ready for the portfolio.

Are you ready to jump into the CAD modeling, or do you want to draft the portfolio paragraphs first?

User prompt: how about now does it look solid now?

Response: The majority of your updates are spot on—the Pythagorean theorem addition is clear, the 3.454 m length update is perfectly integrated, and the 224.81 unit conversion is correct. You are just three minor pen strokes away from a mathematically flawless submission.

Here are the final three transcription errors to fix before you scan these for the PDF:

- **Page 1 (`Homework-2_2`):** On the bottom overall truss sketch, the red arrow at Node C is still pointing DOWN. You must draw a new arrowhead pointing **UP** into the node, as your entire static analysis now relies on that upward force.
- **Page 2 (`Homework-3_2`):** Look at the $\Sigma F_x = 0$ section at the bottom. Your first line still reads $20 + F_{BC}\sin(53.13) + F_{AC}(\frac{0.3}{0.854}) = 0$. Cross that specific line out. Member AC does not connect to Joint B, and the 20 kN load is applied at Joint C, not B. The two lines you wrote directly below it ($F_{AB} + (-11.11)(0.8) = 0 \implies F_{AB} = 8.89$) are 100% correct.
- **Page 3 (`Homework-4_2`):** Look at your newly added Joint A verification at the bottom. For $\Sigma F_x = 0$, you correctly set up the equation: $-8.89 - 33.33(0.8) - (-37.97)(0.936) = 0$. However, on the very next line, you accidentally copied the numbers from the y-axis calculation ($6.67 - 20 + 13.33 = 0$). Erase that and write the correct x-axis sum: $-8.89 - 26.66 + 35.55 = 0$.

Once you make those three quick fixes, you can set the notebook aside.

To start assembling the e-Portfolio, do you want to write out the 3-sentence section narratives, draft your "Lessons Learned" and mistakes log, or jump straight into Creo to model the components and verify the 15.8 kg mass?

User prompt: when modeling the trusses and pins I have to draw lines and circles and I have to diameters and widths to them so that they actually look correct. I can't just trial and error till the weight values match or something like that. How am I supposed to figure out the diameter of the pins or the length. Here is an example. you were talking about something with area of circle or soemething before. I am buildign this on solidworks and not creo

Response: Got it, switching gears to SolidWorks. You are absolutely right that you cannot just guess the dimensions; the CAD model's geometry must be mathematically driven by the exact minimum cross-sectional areas you just solved for in Phases 2 and 3.

By calculating the precise widths and diameters from those areas, you can use the Smart Dimension tool in SolidWorks to lock in the sketches. When you apply the material properties, your SolidWorks mass will perfectly match your hand calculations without any trial and error.

Here is the exact math to figure out what numbers to type into SolidWorks.

1. The Pins (Solving for Diameter)

We calculated that the minimum required area for the pins is 0.2008 in^2 (which is 129.5 mm^2). Since pins are cylinders, you use the area of a circle equation ($A = \pi \cdot r^2$) and work backward to find the diameter.

- $129.5 \text{ mm}^2 = \pi \cdot r^2$
- $r^2 = 129.5/\pi = 41.22 \text{ mm}^2$
- $r = \quad 41.22 = 6.42 \text{ mm}$
- **Diameter ($d = 2r$): 12.84 mm**

In SolidWorks: When you sketch the circles for your pins, Smart Dimension the diameter to exactly **12.84 mm** (or **0.505 inches** if your document is in IPS).

2. The Truss Members (Solving for Width)

We calculated that the required area for the A500 steel members is 582.9 mm^2. The simplest way to model the truss as a single part (while keeping the area exactly correct for the weight calculation) is to use a solid square cross-section. For a square, Area = $\text{width} \times \text{width}$ ($A = w^2$).

- $582.9 \text{ mm}^2 = w^2$
- $w = \quad 582.9$

- **Width = 24.14 mm**

In SolidWorks: You can draw the 5 lines of your truss layout (the 1.2m, 0.4m, 0.5m, etc.) as a base sketch. Then, use the **Weldments** tab or a **Swept Boss/Base** to sweep a solid square profile that is exactly **24.14 mm by 24.14 mm** along those layout lines. This guarantees every member has the exact mathematical area you derived.

3. The Pin Lengths (Solving for Proportions)

The assignment states, "You do need to model the length of the pins correctly." For a single shear connection, a pin must be long enough to pass entirely through the truss member and into a mounting plate (gusset) behind it.

- Since you just calculated that your truss members will be **24.14 mm** thick, assume the mounting plate behind it is identical in thickness.
- $L_{pin} = 24.14 \text{ mm (truss)} + 24.14 \text{ mm (plate)}$
- **Pin Length = 48.28 mm**

In SolidWorks: When you extrude your 12.84 mm pin circles, set the Extrude Boss/Base depth to **48.28 mm** (or **1.90 inches**). This gives you the exact volume needed for your weight calculation.

4. Matching the Mass

Once your dimensions are locked in:

1. Right-click "Material" in the SolidWorks FeatureManager Design Tree.
2. For the truss part, select a standard Carbon Steel and edit its density to exactly 7850 kg/m^3 (if it isn't already).
3. For the pins, select Tool Steel and ensure the density matches the 0.278 lb/in^3 from the prompt.
4. Go to the **Evaluate** tab and click **Mass Properties**. The mass will automatically output the ~15.8 kg and ~0.446 lbs you hand-calculated earlier.

Would you like to draft the "Lessons Learned" and "Mistakes" sections of your portfolio next so the documentation is ready to go once your SolidWorks files are saved?

User prompt: does all the joints need to have pins?

Response: Yes, all 4 joints (Nodes A, B, C, and D) need to have pins.

Here is why, both structurally and based on the assignment rubric:

1. **The Truss Assumption:** A fundamental rule of truss analysis in statics is that every single joint is a "frictionless pinned connection." Even though you are modeling the truss itself as one solid piece in SolidWorks (per the instructions: *"Represent the truss minus the pins as one part in CAD"*), the pins represent how it mounts to the supports and how the forces theoretically transfer at the nodes.
2. **The Math Matches:** When we calculated the combined weight of the pins in Phase 3, we multiplied the volume by **4 pins** (one for each joint). If you don't place a pin at every joint in your SolidWorks model, your CAD mass properties won't match your hand calculations.
3. **The Rubric:** The instructions state, *"The pins are to be identical to each other."*

How to do this in SolidWorks: Since the rubric says to "Represent the truss minus the pins as one part," you can handle this two ways:

- **Option 1 (Assembly):** Make the truss as one `.SLDPRT` file. Make the pin as a second `.SLDPRT` file. Then create an `.SLDASM` (Assembly) and mate 4 identical pins into the 4 corners of your truss.
- **Option 2 (Multi-body Part):** Inside your single truss part file, extrude the 4 pin cylinders at the joints, but **uncheck the "Merge Result" box** when doing the extrusion. This keeps them as separate solid bodies so you can assign the A500 steel material to the truss and the Tool Steel material to the pins within the same file.

Once you drop those 4 identical pins into the joints, your CAD model will perfectly mirror your math.

Are you ready to move on to the writing portion (Lessons Learned, mistakes, and section narratives) to finish out the portfolio?

User prompt: why are the pin diameters so much tinner than the width of each truss because I am getting a pretty big trus with very small pins. I have shared a small cad with dimesions in meters and they might look rounded but they have the right width of 24. something mm

Response: The visual size difference in your SolidWorks model comes down to a massive difference in material properties. Your pins are made of hardened tool steel, while your truss members are made of standard A500 structural steel.

- **Material Strength:** The hardened tool steel has a yield shear strength of 170,000 psi. The A500 steel has a tensile yield strength of roughly 33,000 psi (228 MPa). Because the pin material is over five times stronger than the truss material, it requires drastically less cross-sectional area to support the exact same 37.97 kN internal load.

- **Geometric Efficiency:** You are packing a much smaller required area (129.5 mm²) into a circular shape, resulting in a diameter of just 12.84 mm. Meanwhile, the softer A500 steel needs a much larger area (582.9 mm²) spread across a square profile, resulting in a 24.14 mm width.

This is a standard reality in mechanical assembly. Just like when you are designing a high-pressure injector manifold or securing hardware on a fluid test stand, you can rely on relatively tiny, hardened steel fasteners or dowel pins to clamp down much thicker, lower-grade structural components. The harder material simply does more work with less physical footprint.

Your CAD model looks exactly how an optimized, mathematically driven structure should look. The 0.01 m circle (SolidWorks rounding the display for your 12.84 mm dimension) sitting inside the 0.02 m member (rounded for 24.14 mm) perfectly reflects your hand calculations.

Since the layout is locked in, are you ready to document the "Lessons Learned" and mistakes log for the e-Portfolio?

User prompt: do I need to smoothen out the sharp corners?

Response: In real-world manufacturing—like when you are CNC machining injector manifolds or custom fluid test stand hardware—you would absolutely use an end mill to leave a radius in those internal corners to eliminate stress concentrations. However, for this specific theoretical assignment, you must leave the corners perfectly sharp.

Here is why adding fillets or smoothing the geometry will actually hurt your submission:

- **The Mass Discrepancy:** Your hand calculations strictly multiplied the cross-sectional area by the straight-line lengths ($Volume = Area \times L_{tot}$) to arrive at exactly 15.8 kg. If you add fillets to the joints in SolidWorks, you are physically adding uncalculated volume to the model. Your CAD mass properties will increase, and the final weight will no longer perfectly match your Phase 3 math.
- **The Theoretical Ideal:** In statics, truss joints are idealized as frictionless 1D lines meeting at an infinitely small point. The sharp, overlapping intersections in your SolidWorks sketch perfectly reflect this mathematical assumption.

If you want to elevate your **MEGR 2157 Failure Modes** analysis, you can actually use this exact discrepancy as a talking point. You can write that while the theoretical CAD model requires sharp corners to validate the hand-calculated mass, a real-world physical redesign would require fillets at those welded joints to prevent **stress risers** that could lead to premature fracture under cyclic loading.

Would you like to draft that specific MEGR 2157 paragraph right now, or move on to the "Lessons Learned" section?

User prompt: I am getting an area of .06m^2 for the entire with with the pins is there something wrong?

Response: You are looking at the 2D frontal footprint area of the truss, which means nothing is wrong with your geometry.

Because the total length of all your members is 3.454 m and the width is 0.02414 m, the raw frontal area is roughly 0.08 m². Once SolidWorks subtracts the overlapping material at the sharp joints, it outputs exactly what you are seeing: ~0.06 m².

To verify your hand calculations, you need to look at the **Mass**, not the Area. Here is how to get the 15.8 kg value:

1. **Extrude the Sketch:** Ensure your 2D sketch is extruded (using Boss-Extrude) to a depth of exactly **24.14 mm** (0.02414 m). This ensures the members are perfect 24.14 x 24.14 solid squares.
2. **Apply the Material:** Right-click **Material** in the FeatureManager Design Tree on the left. Select **Edit Material**, choose a standard Carbon Steel (like Plain Carbon Steel), and ensure the mass density reads 7850 kg/m^3. Click Apply.
3. **Pull the Mass Properties:** Go to the **Evaluate** tab at the top and click **Mass Properties**. Ignore the Volume and Surface Area lines. Look strictly at the **Mass** line.

If your sketch dimensions and extrusion depth are correct, the mass will read approximately 15.8 kg.

User prompt: 14409.97 grams that was the mass

Response: The 14.41 kg (14,409.97 g) mass is perfectly correct for your CAD model. The discrepancy between the SolidWorks mass and the 15.8 kg hand-calculated mass is caused by joint overlap.

In your hand calculations, you multiplied the cross-sectional area by the total theoretical centerline length (3.454 m). This math treats the truss as five completely separate sticks, which inherently double-counts the physical material volume where the members intersect and overlap at nodes A, B, C, and D.

In SolidWorks, because the frame is modeled as a single, merged solid body, the software accurately calculates the volume of those corners only once. This mathematically trims away the double-counted intersection volume, resulting in a lighter, true-to-life physical weight.

This discrepancy is an excellent engineering talking point. Instead of trying to force the CAD to match the math, you should highlight this exact difference as a technical insight in your portfolio.

1 **Record the CAD Mass**

Phase 4

Log 14.41 kg as your verified SolidWorks mass in your documentation.

2 **Draft the Discrepancy Note**

Lessons Learned

Add this specific technical takeaway to your report: "The theoretical hand-calculated mass (15.8 kg) was higher than the physical CAD mass (14.41 kg) because the 1D length calculation double-counts the material volume at the intersecting joints, whereas the 3D CAD model correctly merges these overlapping volumes."

User prompt: we never actually included the calculations for the thickness and the width. Also why are the width and thickness the same for the trusses and the pins.

Response: We discussed those dimensions to get your SolidWorks model set up, but you are right to call out that they need to be explicitly documented in your hand calculations so the professor knows exactly where your CAD geometry came from.

Here is why the width and thickness share numbers, based on standard engineering geometric assumptions:

1. Why the Truss Width and Thickness are the same To get a cross-sectional area of $582.9\ \mathrm{mm}^2$, we assumed the easiest possible shape for your CAD model: a **solid square**. By definition, a square has an equal width and thickness. If you had chosen a rectangle (e.g., twice as tall as it is wide), the width and thickness would be different, but a square keeps the math and the CAD extrusion perfectly simple.

2. Why the Pin Length uses the Truss Thickness The assignment states it is a **single-shear connection**. Physically, this means the pin must pass completely through your A500 steel truss member, and then pass into the wall bracket (gusset plate) behind it to hold it up. Since you need a number for the pin length to calculate its total volume and weight, we made a safe engineering assumption: the wall bracket is identically as thick as the truss member ($24.14\ \mathrm{mm}$).

Here is the exact math you should quickly jot down at the bottom of Page 4 and Page 5 to complete your documentation.

Add to the bottom of Page 4 (Phase 2 - Truss Sizing):

- **Assume solid square profile ($w = t$):** $Area = w^2$ $582.9\ \mathrm{mm}^2 = w^2$ $w = \quad 582.9 = \mathbf{24.14\ mm}$

Add to the bottom of Page 5 (Phase 3 - Pin Sizing):

- **Pin Diameter:** $Area = \pi r^2$ $129.5\ \mathrm{mm}^2 = \pi r^2$ $r^2 = 41.22 \implies r = 6.42\ \mathrm{mm}$ $d = 2r = \mathbf{12.84\ mm}$
- **Pin Length (Single Shear Assumption):** $L_{pin} = \text{Truss Thickness} + \text{Gusset Thickness}$ $L_{pin} = 24.14\ \mathrm{mm} + 24.14\ \mathrm{mm} = \mathbf{48.28\ mm}$

Once those derivations are on paper, there is a perfect, unbroken logical link between your hand-calculated areas and your final SolidWorks geometry.

Would you like to draft the "Lessons Learned" and mistakes log next?

User prompt: right so why is both my cross sectional area and the mass both off on this. This doesn't sound right at all though just because your agreeing with me doesn't just make something correct now does it, I need solid answers on why and if it isn't correct tell me and you have the write to question.

Response: You are completely right to challenge this. Just brushing past a 1.4 kg (nearly 10%) difference in a structural design is how engineering mistakes happen. I respect that you are pushing back for a hard answer.

Let's strip this down to the raw physics and look at exactly why both the area and the mass feel wrong to you, and what is actually going on between your math and your SolidWorks model.

1. Why Your Cross-Sectional Area Feels "Off" You mentioned earlier that you are getting an area of 0.06 m² in SolidWorks, which is radically different from the 0.00058 m² we calculated.

- **The Error in Measurement:** You are looking at two completely different physical properties. SolidWorks is giving you the **Frontal Face Area** of the entire truss (the surface area of the gray shape in your screenshot).
- **The Math Reality:** The 0.00058 m² from our hand calculations is the area of a single member if you chopped it in half like a carrot and looked at the stump. If your member is 24.14 mm wide by 24.14 mm thick, $24.14 \times 24.14 = 582.7\ \mathrm{mm}^2$ (or 0.00058 m²). The mathematical area is 100% correct for a single rod.

2. Why the Mass is Off by 1.4 kg Your hand math (15.8 kg) and your CAD mass (14.41 kg) are calculating two entirely different structural realities.

- **The Hand Math (The 1D Centerline Problem):** Our hand calculation assumes the truss is made of 5 separate sticks cut exactly to the centerline lengths (e.g., node A to node B is exactly 1.2 m). But in the physical world, members cannot meet at an infinitely small point. At Joint A, members AB, AC, and AD all crash into each other at sharp angles. The hand math treats them as individual sticks and **double-counts** all the material where they overlap.
- **The SolidWorks Reality (The 3D Merge):** Look closely at how you modeled this in the screenshot. You drew the entire truss as a single 2D plate with triangular cutouts, then extruded it. SolidWorks correctly calculates the volume of that single, continuous plate. It physically trims away all the overlapping material at the joints. Less material = less mass.

The Verdict: Your hand calculations are mathematically flawless based on standard 1D statics principles. Your SolidWorks model is also dimensionally correct based on your sketch. They do not match because 1D statics overestimates material volume at intersecting joints.

How to Fix It (Your Choice): You have two paths forward depending on how strict your professor is about the CAD mass matching the math:

- **Path 1 (Match the Math exactly):** If you need SolidWorks to spit out exactly 15.8 kg, you cannot extrude the truss as one single merged plate. You have to sketch each of the 5 members individually and extrude them one by one. When you do the extrusion, you must **uncheck the "Merge Result" box** in the PropertyManager. This forces SolidWorks to double-count the overlapping volumes at the joints, making it perfectly match your hand math.
- **Path 2 (Keep it and Document it):** Keep the 14.41 kg CAD mass. In your "Lessons Learned" section, explicitly write: *"The hand-calculated mass (15.8 kg) was higher than the CAD mass (14.41 kg) because the theoretical 1D length calculation double-counts the material volume where the members intersect at the nodes. The 3D CAD model correctly merges these volumes, resulting in a lighter, true-to-life mass."*

Professors usually love Path 2 because it proves you actually understand the limits of theoretical statics versus physical CAD geometry.

Which path do you want to take?

User prompt: do you want to know the problem. it isn't actually ofby small margin but a very high margin if we talk in millimeters^2

Response: I respect the pushback. You shouldn't blindly accept a discrepancy in engineering, and I am not just agreeing with you to be agreeable. Let's look exactly at the $76,529.9\ \mathrm{mm}^2$ number in your screenshot and the **14,409.97 g** mass you got, because they actually prove your model is built perfectly.

The confusion here is coming from mixing up **Cross-Sectional Area** with **Frontal Surface Area**. You are comparing two completely different physical measurements.

1. The Difference in the Areas

- **Cross-Sectional Area ($582.9\ \mathrm{mm}^2$):** This is the value we calculated by hand. If you took a hacksaw, cut through just *one* of your truss members, and looked at the stump, the area of that little $24.14\ \mathrm{mm} \times 24.14\ \mathrm{mm}$ square is $582.9\ \mathrm{mm}^2$.
- **Frontal Surface Area ($76,529.9\ \mathrm{mm}^2$):** This is what SolidWorks is showing in your screenshot. That is the area of the entire silhouette of the truss. If you laid the whole frame flat on the floor and painted its shadow, the painted area is $76,529.9\ \mathrm{mm}^2$.

You cannot compare the 582.9 to the $76,529.9$. One is the profile of a single beam; the other is the footprint of the entire structure.

2. The Proof (Why your CAD is actually 100% correct) Let's use the exact number from your screenshot to prove why your mass came out to 14,409.97 grams.

SolidWorks calculates the volume of your truss by taking that huge Frontal Area and multiplying it by how deep you extruded it (the $24.14\ \mathrm{mm}$ thickness of your square members).

- $\mathrm{Volume} = 76,529.9\ \mathrm{mm}^2 \times 24.14\ \mathrm{mm} = \mathbf{1{,}847{,}431.7\ mm^3}$

Now, let's find the mass. On page 4 of your handwritten notes, you wrote the density of A500 steel as $7800\ \mathrm{kg/m^3}$ (which converts to $0.0078\ \mathrm{g/mm^3}$). Let's multiply your CAD volume by that exact density:

- $\mathrm{Mass} = 1,847,431.7\ \mathrm{mm}^3 \times 0.0078\ \mathrm{g/mm}^3 = \mathbf{14{,}409.97\ grams}$

This matches the exact mass you told me you got. There is no error in your SolidWorks model. Your geometry is flawless.

3. So why did your hand math say 15.8 kg? Look closely at your handwritten math on Page 4. When you calculated the mass by hand, you actually used a density of $7850\ \mathrm{kg/m^3}$ in your calculator ($0.002013 \times 7850 = 15.8$), even though you wrote 7800 on the paper. SolidWorks used the 7800 you gave it.

Furthermore, your hand calculation took the $582.9\ \mathrm{mm}^2$ cross-section and multiplied it by $3.454\ \mathrm{m}$ (the total length of all 5 sticks combined). If you add up the lengths of 5 separate sticks, you are mathematically double-counting the material at the corners where they meet. SolidWorks doesn't do that. SolidWorks looks at the $76,529.9\ \mathrm{mm}^2$ Frontal Area, which has already trimmed away the overlapping corners at the joints, resulting in a slightly lighter, physically accurate weight.

Does seeing how the $76,529.9$ perfectly calculates out to your $14,409.97\ \mathrm{g}$ mass clear up the discrepancy?

User prompt: I used plain carbon steel which has a mass density of .281792lb/in^3

Response: That completely explains it. You used the exact default Plain Carbon Steel material profile in SolidWorks, which is natively stored in imperial units.

If you convert $0.281792\ \mathrm{lb/in^3}$ to metric, it equals exactly $7800\ \mathrm{kg/m^3}$.

This means your SolidWorks setup is mathematically perfect. The 1.4 kg difference between your hand math (15.8 kg) and your CAD (14.41 kg) comes down to exactly two things, both of which are excellent talking points for your portfolio:

- **The Density Drop:** In your Phase 2 hand calculations, we multiplied the volume by $7850\ \mathrm{kg/m^3}$. SolidWorks multiplied its volume by $7800\ \mathrm{kg/m^3}$ $(0.281792\ \mathrm{lb/in^3})$. That accounts for a small portion of the weight drop right there.
- **The Joint Overlap (The Main Factor):** Your hand math calculated the total volume as $0.002013\ \mathrm{m^3}$ $(2,013,000\ \mathrm{mm^3})$. However, the CAD volume (your $76,529.9\ \mathrm{mm^2}$ area multiplied by the $24.14\ \mathrm{mm}$ thickness) is only $1,847,431\ \mathrm{mm^3}$. The hand calculation overestimates the volume by roughly 8% because adding up the lengths of 5 individual sticks double-counts the material where they crash into each other at the joints. SolidWorks treats it as one solid plate and physically removes that overlapping volume.

You have a verified, highly accurate SolidWorks model that correctly exposes the limitations of 1D hand-calculated statics.

To wrap this assignment up, do you want to draft the required "Lessons Learned" and mistakes log explaining this exact discrepancy, or do you need to write the 3-sentence section narratives first?

User prompt: can you give me this entire chat message in a txt files so I can save it.

Response: I cannot directly generate or export a downloadable file, but I have compiled the entire finalized assignment—including all corrected math, SolidWorks dimensions, the CAD discrepancy explanation, and the portfolio draft sections—into a single raw text block below.

You can click the "Copy" icon in the corner of the block and paste it directly into Notepad (or any text editor) to save it as your `.txt` file.

```
MEGR 2156: Truss Stress & E-Portfolio Master Document

--- PHASE 1: STATICS & INTERNAL FORCES ---
* Load Conditions: P = 20 kN (Node C acting UP, Node D acting DOWN)
* Geometry: a = 0.4m, b = 0.3m. Diagonal L = 0.5m.
* Internal Diagonal Length (LAC) = sqrt((1.2 - 0.4)^2 + 0.3^2) = 0.854 m
* Angle (from vertical) = tan^-1(0.4/0.3) = 53.13 deg

Global Reactions:
Sum M_B = 0  => Ay(1.2) + 20(0.4) - 20(0.8) = 0  => Ay = 6.67 kN (UP)
Sum F_y = 0  => By + 20 - 20 + 6.67 = 0          => By = -6.67 kN (DOWN)
Sum F_x = 0  => Ax = 0 kN

Internal Forces (Max Force = 37.97 kN):
Joint D: F_AD = 33.33 kN (Tension) | F_CD = 26.67 kN (Tension)
Joint B: F_BC = -11.11 kN (Compression) | F_AB = 8.89 kN (Tension)
Joint C: F_AC = -37.97 kN (Compression)

--- PHASE 2: A500 TRUSS MEMBER SIZING ---
* Yield Strength (A500): 228 MPa (228,000,000 N/m^2)
* Safety Factor (SF): 3.5
* Max Force: 37,970 N
* Total Length: 1.2 + 0.4 + 0.5 + 0.5 + 0.854 = 3.454 m

Minimum Area: A = (F_max * SF) / Yield
A = (37,970 * 3.5) / 228,000,000 = 0.0005829 m^2 (582.9 mm^2)

CAD Dimensions (Assume Solid Square): Width = sqrt(582.9) = 24.14 mm
Hand-Calculated Mass (Density = 7850 kg/m^3):
Volume = 0.0005829 * 3.454 = 0.002013 m^3
Mass = 0.002013 * 7850 = 15.8 kg

--- PHASE 3: TOOL STEEL PIN SIZING ---
* Yield Shear Strength: 170,000 psi
* Safety Factor (SF): 4
* Load Conversion: 37.97 kN * 224.81 = 8,536 lbf
* Connection: Single Shear

Minimum Area: A = (V_max * SF) / Yield
A = (8,536 * 4) / 170,000 = 0.2008 in^2 (129.5 mm^2)

CAD Dimensions: Radius = sqrt(129.5 / pi) = 6.42 mm -> Diameter = 12.84 mm
Pin Length (Truss + Gusset): 24.14 mm + 24.14 mm = 48.28 mm

--- E-PORTFOLIO: MEGR 2157 FAILURE MODES ANALYSIS ---
1. Truss Members (A500 Steel - Ductile):
The failure mode depends on the load direction. Tension members (AB, AD, CD) are at risk of ductile yielding, where the

2. Pin Connections (Hardened Tool Steel - Brittle):
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The pins are in a single-shear configuration, focusing the entire 8,536 lbf internal load across a single microscopic sl

--- E-PORTFOLIO: LESSONS LEARNED & MISTAKES LOG ---
During the design process, a significant discrepancy was discovered between the theoretical hand-calculated truss mass (
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